FACULTY OF COMPUTING
UTM Johor Bahru

SECD2613
SYSTEM ANALYSIS AND DESIGN
Semester 02 2025/2026

PROPOSAL

Title :
UTMShop: A Smart E-Commerce System Designed for UTM Community to Buy, Sell, and Trade Within Campus

Group : 4

Team Members :

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PROPOSAL **1**
1. INTRODUCTION **3**
2. BACKGROUND STUDY **4**
3. PROBLEM STATEMENT **5**
4. PROPOSED SOLUTION **6**
Feasibility Study 7
1. Technical Feasibility 7
2. Operational Feasibility 7
3. Economic Feasibility (Cost-Benefit Analysis – CBA) 8
5. OBJECTIVES **10**
6. SCOPE PROJECT **11**
6.1 Project Inclusions **11**
6.2 Project Exclusions 11
7. PROJECT PLANNING **12**
7.1 HUMAN RESOURCE 12
7.2 WORK BREAKDOWN STRUCTURE (WBS) 13
7.3 PERT CHART 14
7.4 GANTT CHART 17
8. BENEFIT AND SUMMARY **18**

1. INTRODUCTION

Nowadays, online platforms play an important role in making buying and selling easier. However, students and staff at Universiti Teknologi Malaysia (UTM) still rely on platforms such as WhatsApp or Telegram groups to carry out transactions. These platforms are often unorganized, making it difficult for users to search for items, communicate effectively, and manage transactions smoothly.

As a result, users may face several problems such as difficulty in finding the items they need, unclear communication, and delays in the buying and selling process. This can cause inconvenience and make the overall experience less efficient.

To solve these issues, UTMShop is proposed as an online marketplace specially designed for the UTM community. The system provides a more organized and user-friendly platform where students and staff can buy and sell items easily within the campus. Since it focuses only on UTM users, it also creates a safer and more trusted environment for transactions.

This project aims to develop a platform that improves convenience and makes the buying and selling process within the UTM community more efficient.

2. BACKGROUND STUDY

The modern world has seen significant changes in the way business is conducted in recent years. With the rapid development of technology and the exponential rise of e-commerce, traditional markets have been replaced by digital ones, creating an entire "on-demand" culture among consumers. Being in an environment that requires a high level of efficiency and productivity at **Universiti Teknologi Malaysia (UTM)**, the need for a digital marketplace that would facilitate transactions becomes particularly relevant.

As is the case with many other universities, the UTM community has created its own micro-economy, which is self-supportive and thriving thanks to student-led initiatives. Student businesses sell everything from homemade food to second-hand books. Moreover, students actively use each other's services when it comes to, for example, getting documents printed, carpooling, tutoring, and others. These practices greatly contribute to the prosperity of students' lives and make life easier and more convenient at a university.

However, while the amount of transactions being made is enormous, the current campus market lacks organization and centralization. At present, all commercial activity takes place in an ad-hoc manner using various communication tools such as WhatsApp groups, Telegram posts, and social media. This creates significant difficulties and problems for sellers, who struggle with low visibility and the tedious task of manually managing orders through endless chat logs. Likewise, buyers face challenges in navigating these fragmented channels, often leading to missed opportunities and a lack of trust in transaction security.

Consequently, there is a clear demand for a **unified e-commerce platform** tailored specifically for the UTM community. By consolidating these scattered products and services including a dedicated system for campus food delivery into a single, organized interface, the proposed platform will streamline the campus trading experience. This initiative aims to bridge the gap between student entrepreneurs and the wider university committee, fostering a more efficient, secure, and integrated digital economy on campus.

3. PROBLEM STATEMENT

Although the campus commerce appears to be rather dynamic, several important factors affect the efficiency and development of the economy managed by the university students themselves. The following problems have been highlighted as the main challenges for the UTM community:

1. Incompatibility of the current platforms.

Nowadays, students use common e-commerce websites, such as Shopee, and private social media groups to purchase their items. However, these platforms were not specifically designed for this type of business since they require relatively high shipping fees and a long waiting period before receiving the products. Besides, such platforms do not allow students to browse through all their choices and cannot help manage the inventories.

2. Absence of the secure trading environment.

In the current situation, trust becomes an issue between participants of online trade because many users hesitate to interact with unknown people from other towns or countries. Since there is no possibility to verify one's identity within the UTM community, students doubt the quality of the product and safety of online payments. Moreover, this lack of verification puts people in the position of vulnerability to possible frauds or other kinds of scams.

3. Fragmented products and services.

There is a notable absence of a unified platform that integrates various campus needs. Currently, a student might have to check a Telegram channel for second-hand books, a WhatsApp group for food delivery, and a separate contact for printing services. This fragmentation makes the purchasing process time-consuming and inefficient, as there is no "one-stop" solution that combines the buying and selling of physical items with essential campus services.

4. Difficulty in accessing essential academic resources.

It is increasingly difficult for students to find affordable and relevant academic materials, such as textbooks, past-year notes, and used laboratory equipment, in an organized manner. Because information is scattered across multiple unindexed channels, these useful items often go unnoticed or are discarded. This lack of a structured marketplace prevents students from easily accessing low-cost resources that are vital for their academic success.

4. PROPOSED SOLUTION

The proposed solution is **UTMShop**, a dedicated **web-based marketplace platform** designed exclusively for the UTM community (students, lecturers, and staff). It serves as a centralized, secure, and user-friendly e-commerce platform that combines buying/selling of second-hand items with on-campus service bookings, eliminating the current reliance on fragmented WhatsApp/Telegram groups and general platforms like Shopee.

Key Features of UTMShop

- **UTM Email Domain Authentication** (@utm.my) – only verified UTM users can register and transact.
- **Product Listings** – books, notes, electronics, furniture, daily essentials, and used goods.
- **Service Listings** – printing, transportation (on-campus), tutoring, food delivery (campus-only), and other student-related services.
- **Advanced Search & Filters** – by category, price range, location (specific hostel/ faculty), condition, and availability.
- **Simple Booking System** for services with real-time slot availability.
- **Rating & Review System** for both products and service providers to build trust.
- **Real-time Chat** between buyers, sellers, and service providers.
- **Payment & Delivery Options** – Cash on Delivery (COD), self-pickup at designated campus points, or on-campus delivery (no off-campus delivery).
- **Vendor Management Dashboard** for sellers to manage listings and orders.

The platform will be developed as a responsive web application (desktop and mobile-friendly) using modern web technologies.

Feasibility Study

1. Technical Feasibility

UTMShop is highly feasible with current technology:

- **Front-end**: HTML5, CSS3, JavaScript (React.js or Bootstrap for responsive UI).
- **Back-end**: PHP/Laravel or Node.js/Express with MySQL database (or Firebase for real-time chat).
- **Authentication**: Integration with UTM's existing email system (simple domain validation + optional SSO if API is available).
- **Hosting**: Can be deployed on UTM's existing server infrastructure or affordable cloud services (Hostinger, Vercel, or UTM's student hosting facilities) at low cost.
- **Security**: HTTPS, data encryption, and role-based access (buyer/seller/admin). Real-time chat can use Socket.io or Firebase.
- **Existing tools**: No new hardware is required beyond standard development laptops. All required software (VS Code, XAMPP, GitHub) is free or already available to students.

Conclusion: Technically feasible. The team possesses the necessary programming skills (web development) and the project can be completed within one semester using open-source tools.

2. Operational Feasibility

- **User Acceptance**: The UTM community (approximately 20,000+ students and staff) already faces the exact problems described in the problem statement. A campus-only platform with UTM email login will significantly increase trust and adoption.
- **Ease of Use**: Simple, intuitive interface similar to Shopee but tailored for campus needs (location filters, self-pickup, no shipping fees).
- **Training & Support**: Minimal training required. A short onboarding tutorial and FAQ will be provided. The platform admin (student team) can provide support via the in-app chat.
- **Organizational Fit**: Fully aligns with UTM's digital transformation goals and supports student entrepreneurship without needing physical warehousing or external logistics.

Conclusion: Operationally feasible. High user adoption is expected because it directly solves daily pain points (messy groups, trust issues, delivery fees) and requires no behavioural change beyond using a new UTM-branded app.

3. Economic Feasibility (Cost-Benefit Analysis – CBA)

Estimated cost and expected benefit for UTMShop

Estimated Costs	
Hardware	RM40000
Software	RM20000
Salary	RM45000
Maintenance	RM4000 per year
Transaction Fees	RM6000
Marketing	RM3000

Assumptions	
Discount Rate	10%
Sensitivity Factor (cost)	0.5
Sensitivity Factor (benefit)	0.7
Annual Change In Production Costs	7%
Annual Change In Benefits	5%

Estimated Benefits	
Saving	RM4000 per month
Increase Sales	RM15000

Calculate CBA for UTMShop

CRITERIA	YEAR			
	0	1	2	3
DEVELOPMENT				
Hardware	20000			
Software	10000			
Salary	22500			
Total Development Cost	52500			

OPERATION				
Maintenance		2000	2140	2290
Transaction Fees		3000	3210	3435
Marketing		1500	1605	1717
Total Operation Cost (Annual)		6500	6955	7442
Present Value (PV)		5909	5748	5591
Accumulated Cost		58409	64157	69748
Saving		33600	35280	37044
Increase Sales		10500	11025	11576
Total Benefit (Annual)		44100	46305	48620
Present Value		40091	38269	36529
Accumulated Benefit		92591	130860	167389
Gain/Loss		86091	123905	159947
Profitability Index (PI)	3.04			

5. OBJECTIVES

- To design and develop a centralized web-based marketplace (UTMShop) by the end of Semester 2, 2025/2026, to replace fragmented WhatsApp and Telegram groups.

- To ensure 100% secure user access by implementing a mandatory UTM email domain (@utm.my) authentication system for all registered students and staff.

- To integrate a real-time communication module within the platform that allows instant chat between buyers, vendors, and admins for efficient order coordination.

- To provide a structured digital catalog featuring at least five distinct categories (books, food, printing services, daily essentials and used goods) to improve product searchability for the UTM community.

- To achieve a high standard of service reliability by incorporating a five-star rating and review system to monitor and evaluate vendor performance.

- To facilitate 100% on-campus flexibility by providing users with clear choices between self-pickup or on-campus delivery options during the checkout process.

6. SCOPE PROJECT

6.1 Project Inclusions

- **User Authentication**: A secure login system restricted to verified UTM students and staff using university email domains (@utm.my).
- **Vendor Management**: Comprehensive tools for campus shops and student entrepreneurs to manage product listings, pricing, and real-time inventory.
- **Digital Catalog**: A structured and searchable database that categorizes goods and services (food, books, printing, and transportation) for easy navigation.
- **Ordering System**: Integrated shopping cart functionality allowing users to select multiple items and submit digital orders systematically.
- **Rating & Review System**: A transparent feedback mechanism that enables users to score and review vendors to maintain campus service quality.
- **Real-time Chat**: An in-platform messaging feature providing direct communication between buyers, vendors, and admins for order coordination.
- **Flexible Fulfillment**: Provides users with the choice between "Self-Pickup" at designated campus locations or "On-Campus Delivery."
- **Admin Dashboard**: A management interface for platform administrators to monitor transactions, verify vendors, and manage user roles.

6.2 Project Exclusions

- **External User Access**: The platform is strictly private, registration and access are not permitted for individuals outside the UTM community.
- **Off-Campus Delivery**: Delivery services are geographically restricted to the UTM Skudai campus boundaries, no external logistics are supported.
- **Physical Warehousing**: The platform acts as a digital intermediary and does not provide physical storage or warehouse space for vendor products.
- **Direct Sales & Inventory**: The system is a "platform-only" model, it does not sell self-owned products or take responsibility for vendor stock replenishment.
- **Packaging Services**: The platform does not provide or subsidize packaging materials such as bags and boxes, this remains the vendor's responsibility.
- **Real Money Transactions**: To ensure security and simplify the scope, the system will only support simulated digital wallets or Cash on Delivery (COD).
- **Native Mobile Application**: The project is strictly a responsive web-based application and does not include the development of iOS or Android apps.

7. PROJECT PLANNING

7.1 HUMAN RESOURCE

The project is executed by **Group 4** (7 members). Roles are assigned based on individual

No.	Student Name	Matric Number	Assigned Role(s)
1	MUHAMMAD ALIF BIN AZMAN	A24CS5042	Project Leader, Backend Developer, Database Design
2	MUHAMMAD FIRDAUS ALIF BIN ROSLAN	A24CS5044	Frontend Developer, UI/UX Design
3	NURUL IMAN BINTI MOHAMAD ZAKRY	A25CS0131	Requirements Analysis, Testing & QA
4	NURUL HAFIZZAH BINTI MOHAMAD RSELI	A25CS0334	System Design, Documentation, Deployment
5	MALEK ALRIHAWI	A25CS4013	Real-time Chat Integration, Vendor Management Module
6	ERIC TAN ZHI EN / SIA JUN YANG	A25CS0058 A25CS0138	Testing & QA, Security Implementation, Support

strengths and aligned with SDLC phases for balanced workload and collaboration. All members will contribute to the GitHub repository.**GitHub Repository & Project Management (as required by Phase 1 guidelines)**

- Repository name: **Group4_Project1_SAD_20252026**
- All team members have been invited as collaborators.
- A **Kanban board** has been created in GitHub Projects with columns: To Do, In Progress, Review, Done.
- Tasks are assigned to individuals and updated weekly.
- Version control best practices (feature branching, pull requests, code reviews) will be followed strictly.
- Repository URL and snapshot will be submitted with the final report.

7.2 WORK BREAKDOWN STRUCTURE (WBS)

The project structure follows a **Process-oriented WBS**, as we are focusing on the sequential progression of our task.

Level 1 :
1.0 UTMShop

Level 2 :
1.1 Project Initiation
1.2 Planning
1.3 Data Collection
1.4 Analyzing Data
1.5 System Design
1.6 Development
1.7 Unit Testing
1.8 Implementation and Evaluation

Level 3
<u>1.1 Project Initiation</u>
1.1.1 Define Scope
1.1.2 Identify Stakeholders
1.1.3 Project Approval

<u>1.2 Planning</u>
1.2.1 Define Budget
1.2.2 Resource Allocation

<u>1.3 Data Collection</u>
1.3.1 Conduct Survey (Google Form)
1.3.2 Conduct JAD Session

<u>1.4 Analyzing Data</u>
1.4.1 Analyze Requirement
1.4.2 Perform Feasibility Study

<u>1.5 System Design</u>
1.5.1 System Architecture
1.5.2 UI Wireframes
1.5.3 Database Scheme

1.6 Development
1.6.1 Backend Development
1.6.2 Frontend Development
1.6.3 Database Setup

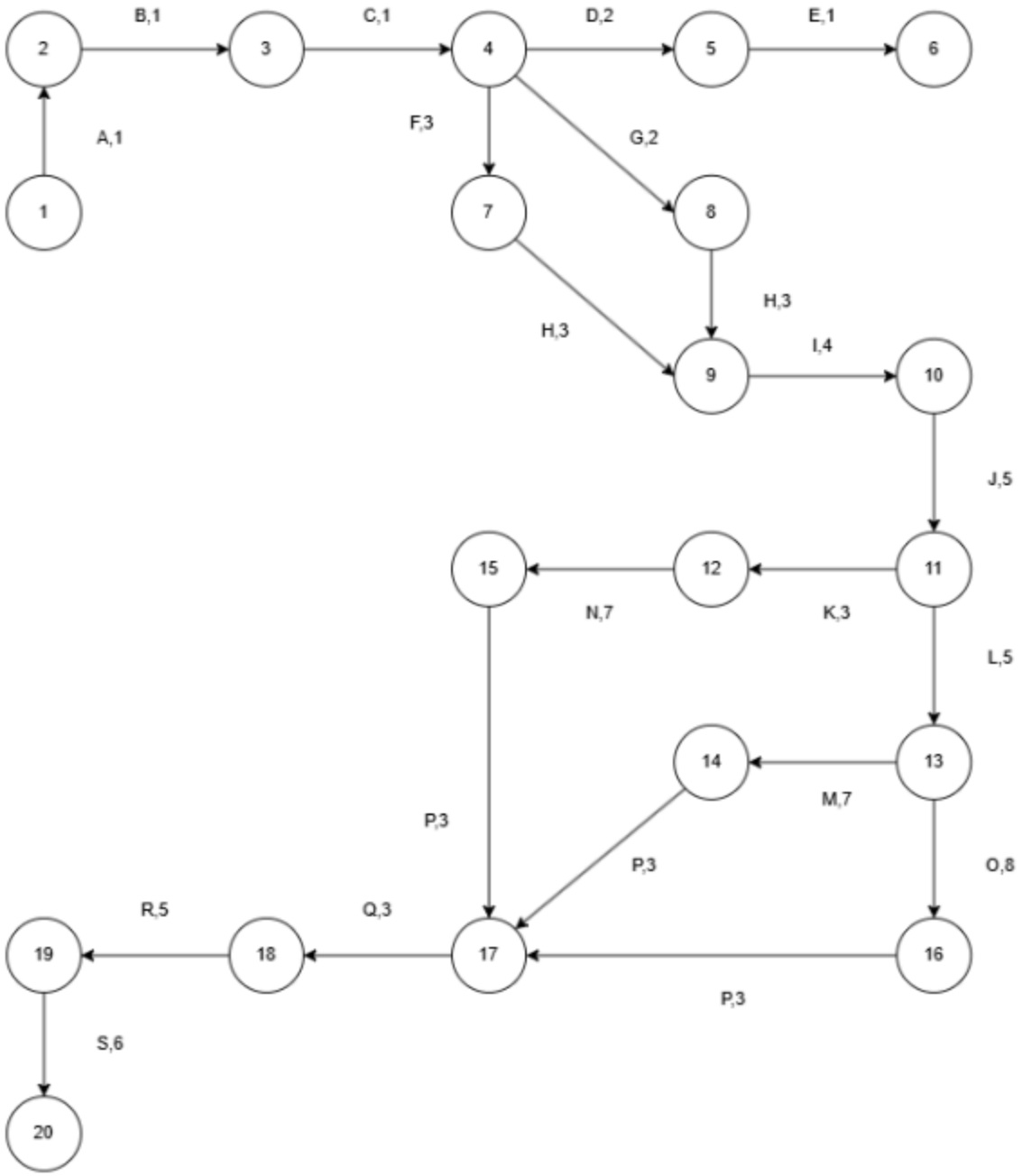
1.7 Testing
1.7.1 Unit Testing
1.7.2 System Testing

1.8 Implementation and Evaluation
1.8.1 System Evaluation
1.8.2 Presentation Preparation

7.3 PERT CHART

ID	Activity	Predecessor	Duration (days)
A	1.1.1 Define Scope	None	1
B	1.1.2 Identify Stakeholders	A	1
C	1.1.3 Project Approval	B	1
D	1.2.1 Define Budget	C	2
E	1.2.2 Resource Allocation	D	1
F	1.3.1 Conduct Survey	C	3
G	1.3.2 Conduct JAD Session	C	1

H	1.4.1 Analyze Requirement	F, G	3
I	1.4.2 Feasibility Study	H	4
J	1.5.1 System Architecture	I	5
K	1.5.2 UI Wireframes	J	3
L	1.5.3 Database Scheme	J	5
M	1.6.1 Backend Development	L	7
N	1.6.2 Frontend Development	K	7
O	1.6.3 Database Setup	L	8
P	1.7.1 Unit Testing	M, N, O	3
Q	1.7.2 System Testing	P	3
R	1.8.1 System Evaluation	Q	5
S	1.8.2 Presentation	R	6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7.4 GANTT CHART

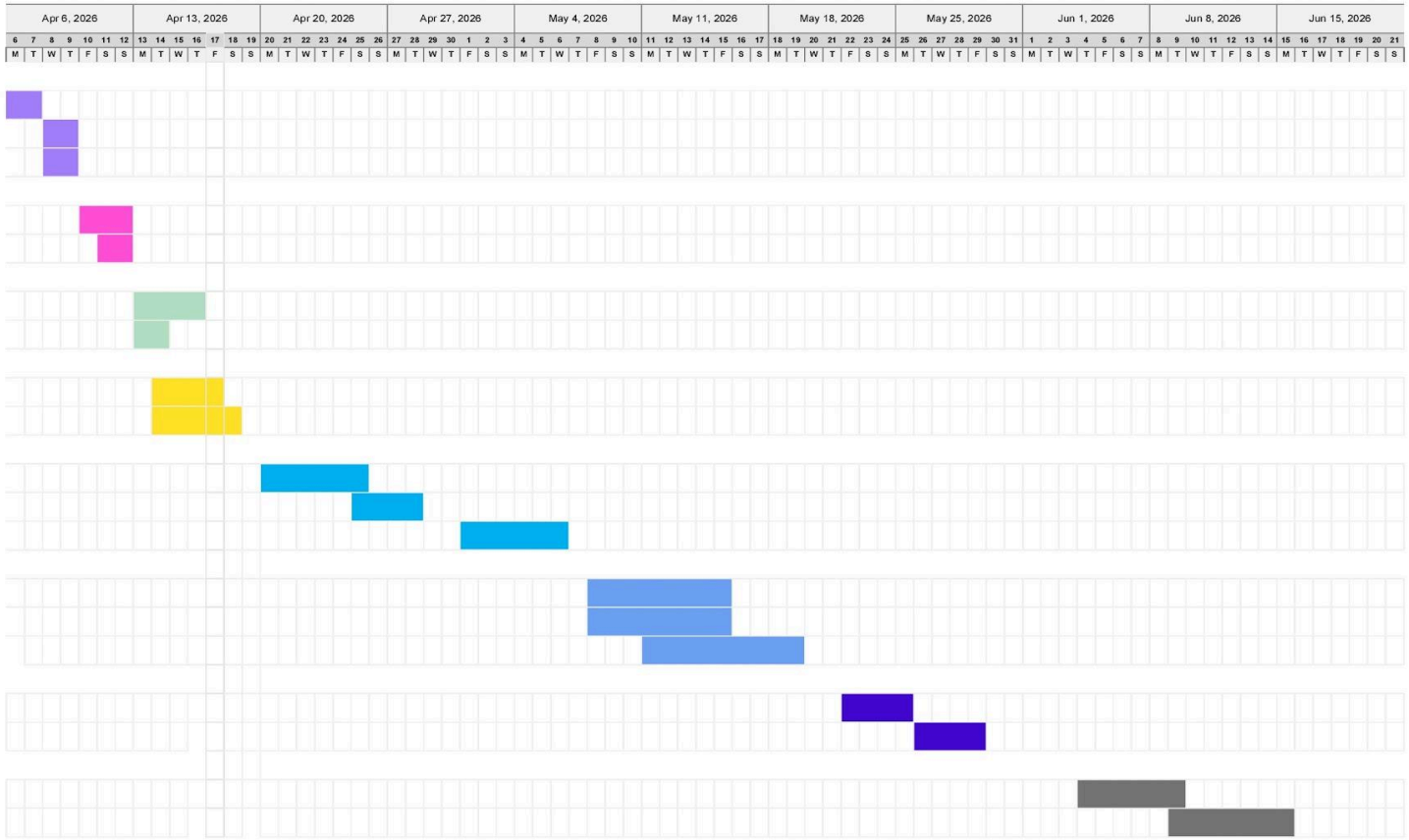
UTMShop

A Smart E-Commerce System Designed for UTM Community to Buy, Sell, and Trade Within Campus

TASK	ASSIGNED TO	START	END	DAYS
Project Initiation				
Define Scope	All members	06-Apr-26	07-Apr-26	1
Identify Skateholders	All members	08-Apr-26	09-Apr-26	1
Project Approval	All members	08-Apr-26	09-Apr-26	1
Planning				
Define Budget	Firdaus	10-Apr-26	12-Apr-26	2
Resource Allocation	All members	11-Apr-26	12-Apr-26	1
Data Collection				
Conduct Survey (Google Form)	All members	13-Apr-26	16-Apr-26	3
Conduct JAD Session	All members	13-Apr-26	14-Apr-26	1
Analyzing Data				
Analyze Requirements	All members	14-Apr-26	17-Apr-26	3
Perform Feasibilty Study	Firdaus	14-Apr-26	18-Apr-26	4
System Design				
System Architecture	All members	20-Apr-26	25-Apr-26	5
UI Wireframes	All members	25-Apr-26	28-Apr-26	3
Database Scheme	All members	01-May-26	06-May-26	5
Development				
Backend Development	All members	08-May-26	15-May-26	7
Frontend Development	All members	08-May-26	15-May-26	7
Database Setup	All members	11-May-26	19-May-26	8
Testing				
Unit Testing	All members	22-May-26	25-May-26	3
System Testing	All members	26-May-26	29-May-26	3
Impelementation and Evaluation				
System Evaluation	All members	04-Jun-26	09-Jun-26	5
Presentation Preparation	All members	09-Jun-26	15-Jun-26	6

Project start: 06-Apr-26

Display week: 1



8. BENEFIT AND SUMMARY

Overall Summary

UTMShop is a secure, campus-exclusive web-based marketplace that directly solves the fragmentation, trust, and cost issues faced by the UTM community. It combines product trading and on-campus services in one easy-to-use platform, built using modern web technologies and managed through professional project planning (WBS, PERT, Gantt, GitHub and Kanban).

Key Benefits

- **Economic Benefits** (from CBA in Section 4.0):
  - Profitability Index = **3.04**
  - Positive gain from Year 1 onwards (RM 86,091 in Year 1)
  - Total accumulated benefit by Year 3 = RM 167,389
  - Breaks even within the first year of operation.
- **Operational Benefits**: One centralized platform replaces messy WhatsApp/Telegram groups and Shopee, faster transactions, better organization, and real-time support.
- **Social Benefits**: Builds trust through UTM authentication and ratings, supports student entrepreneurship and campus community.
- **Strategic Benefits**: Aligns with UTM's digital goals, low maintenance cost, and high user adoption expected from 20,000+ community members.

The project is technically, operationally, and economically feasible. With clear objectives, well-defined scope, detailed project planning, and strong team collaboration, UTMShop will be successfully delivered by the end of the semester.

GitHub Repository (to be submitted):
https://github.com/vynxMY/Group4_Project1_SAD_20252026

(Repository snapshot and Kanban board screenshot will be attached in the final submission.)